CAREERTREE — SYSTEM ARCHITECTURE & PROJECT OVERVIEW (C)

A Deterministic Binary Decision Tree Career Recommendation Engine Implemented in Pure C

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Implementation: Pure C (C99, GCC 15.2.0) | **Repository:** github.com/bshubhayu07/career-tree

1. Executive Summary

CareerTree is an educational career advisory system engineered entirely in **Pure C** around an explicit **Deterministic Binary Decision Tree** data structure. Rather than relying on opaque, black-box machine learning algorithms or flat questionnaires that offer zero visibility into why a recommendation was made, CareerTree grounds career guidance in fundamental Computer Science principles. Built with raw struct pointers (DecisionNode* yes; and DecisionNode* no;), manual heap allocation (malloc), and an auxiliary LIFO stack for backtracking, CareerTree delivers 100% explainable, auditable career guidance in logarithmic **O(log■ N)** time.

2. The Problem: Navigating Modern Tech Specialization

Modern computer science has fractured into specialized disciplines, leaving students confused between overlapping domains:

- **Domain Ambiguity:** Disentangling adjacent disciplines like *AI Research* (theoretical mathematics) vs. *MLOps* (production pipelines).
- **Black-Box Opacity:** Commercial guidance platforms produce scores without an auditable, deterministic decision path.
- **Absence of Backtracking:** Static forms cannot maintain runtime state, preventing users from exploring alternatives.

3. Core Conceptual Model & Tree Halving Principle

CareerTree models career guidance as a balanced binary decision tree of depth 4 ($h = 4$). Each internal node presents a targeted binary question, dividing the remaining search space into mutually exclusive subtrees (**Yes** and **No**). Starting from 16 possible pathways at the root, every decision eliminates exactly half of the remaining candidate space:

Level 0 (Root)	Level 1 Decisions	Level 2 Decisions	Level 3 Decisions	Level 4 (Leaves)
16 Pathways	8 Pathways Left	4 Pathways Left	2 Pathways Left	1 Exact Match Found!

4. Key Architectural Highlights in C

- **Raw Memory Pointers:** Explicit struct pointers (yes and no) without high-level runtime abstractions.
- **Logarithmic Search:** Evaluates precisely 4 sequential binary choices to isolate a recommendation in $O(h) = 4$ steps.
- **LIFO Stack Backtracking:** Custom stack pops visited nodes in $\mathcal{O}(1)$ time to undo decisions cleanly.
- **Zero Memory Leaks:** Post-order recursive destructor (free_tree) frees all 31 nodes and payloads on heap.

Technology Stack & Architectural Utilization in C

Detailed technical breakdown of C components, memory allocation, and compilation toolchain.

1. C99 Standard & The GCC 15.2.0 Toolchain

CareerTree is implemented strictly in **C99** (conforming to ISO/IEC 9899:1999) and compiles with GCC using -Wall -Wextra -O2. Building in pure C guarantees zero runtime dependencies, direct hardware memory control, minimal executable footprint (< 150 KB), and sub-millisecond execution speeds. The program compiles cleanly via an automated Makefile.

2. Modular Multi-File Architecture

The codebase is divided into three distinct modules to ensure separation of concerns and maintainability:

- **tree.h (Header Interface):** Defines struct DecisionNode, struct CareerProfile, struct NodeStack, and struct Queue, declaring all public prototypes and constants.
- **tree.c (Algorithmic Engine):** Implements heap allocation (malloc), bottom-up tree builder (31 nodes), BFS FIFO queue, DFS recursive search with backtracking, and recursive memory destructor.
- **main.c (CLI Interface & Tests):** Implements interactive terminal UI with ANSI colors, LIFO undo stack, ASCII tree graph renderer, file export, and built-in unit tests.

3. Memory Management & Heap Safety

Every node and career profile is dynamically allocated on the heap via malloc() with defensive NULL checks. When the application terminates (Menu Option 7), free_tree(root) executes a recursive post-order traversal freeing both subtrees and the node payload before freeing the node itself, guaranteeing 100% memory reclamation.

4. Technology & Component Mapping Matrix

Component / Tool	Role in System	Specific Implementation	Value Delivered
C99 (GCC 15.2.0)	Core Language	tree.c, main.c, tree.h	Raw pointer manipulation, manual heap allocation, direct hardware c
struct DecisionNode*	Tree Node Modeling	tree.h, tree.c	Explicit left/right pointer dereferencing for O(log2 N) search
struct NodeStack	LIFO Stack (Backtrack)	tree.h, main.c	Enables O(1) undo and history rollback during assessment
struct Queue (Linked)	FIFO Queue (BFS)	tree.h, tree.c	Implements Breadth-First Search level-order traversal
GNU Makefile	Build Automation	Makefile	Automated compilation with -Wall -Wextra -O2 flags
File I/O (fopen)	Dossier Export	tree.c (export_career_dossier)	Serializes complete assessment report to career_recommendation.t
Built-in C Unit Tests	Quality Assurance	main.c (run_unit_tests)	Validates 100% path coverage across all 16 permutations in 0.000s

Data Structures, Algorithms & Traversal Mechanics

Memory layout of C structs, pointer mechanics, stack/queue operations, and complexity analysis.

1. C Struct Memory Layout

Nodes are defined with explicit pointer members in tree.h:

```
typedef struct DecisionNode { char question[512]; // Decision question string char context_tip[512]; // Contextual evaluation tip struct DecisionNode* yes; // Pointer to Left child (YES) struct DecisionNode* no; // Pointer to Right child (NO) CareerProfile* career; // Pointer to leaf payload (or NULL) } DecisionNode;
```

2. Mathematical Properties of the Binary Tree

The C implementation satisfies all formal properties of a **Full and Perfectly Balanced Binary Tree**:

Property Metric	Mathematical Formula	Computed Value	System Implication
Tree Height (h)	h = depth of leaves	4 levels	Guarantees exactly 4 questions per user session
Total Nodes (N)	N = 2^(h+1) - 1	31 nodes	15 internal decision nodes + 16 leaf outcomes
Leaf Count (L)	L = 2^h	16 leaves	16 unique, mutually exclusive tech career trajectories
Internal Nodes (I)	I = 2^h - 1	15 nodes	15 strategic branching tests dividing the problem space
Balance Factor	|h_left - h_right|	0 (Perfect)	Guarantees optimal logarithmic search efficiency O(log■ N)

3. Traversal & Backtracking Algorithms in C

• **Pointer Dereferencing Traversal:** At each step, user input determines current = (choice == 'Y') ? current->yes : current->no;. Total traversal completes in exactly $h = 4$ pointer dereferences (< 0.05 milliseconds).
• **LIFO Stack Backtracking:** Visited nodes are pushed to NodeStack. Clicking 'Undo' invokes stack_pop(), reverting to the parent node in $\mathcal{O}(1)$ time without memory leaks.
• **Recursive DFS Search:** find_path_to_career() searches the tree for any target career title. If a branch fails, it backtracks by decrementing (*path_len)--.
• **Linked Queue Level-Order BFS:** print_bfs_traversal() uses dynamic linked queue nodes to visit all nodes layer-by-layer.

4. Asymptotic Complexity Summary

Operation	Data Structure / Algorithm	Time Complexity	Space Complexity
Inference Traversal	Direct pointer dereferencing	O(h) = O(log■ N) = 4 steps	O(h) stack history
Backtrack (Undo)	LIFO stack pop operation	O(1) constant time	O(1) auxiliary
Path Search	Recursive DFS with backtracking	O(N) worst-case	O(h) recursion depth
Level-Order Scan	FIFO linked queue (BFS)	O(N) full tree scan	O(L) = O(16) queue size
Memory Cleanup	Recursive post-order traversal	O(N) all nodes freed	O(h) recursion stack

Real-World Applications, Industry Impact & Results

Practical deployment domains, taxonomy of the 16 career outcomes, and user journey workflow.

1. Real-World Application Domains

CareerTree is engineered as a high-performance decision-support engine across academic and corporate environments:

- **Undergraduate Academic Advising:** Deployed during freshman orientation to guide engineering students toward majors, minors, and elective tracks based on objective technical affinities.
- **EdTech & Coding Bootcamps:** Serves as an automated onboarding diagnostic for online learning platforms (Coursera, bootcamps), routing learners to appropriate curriculum tracks (e.g. Web Dev vs AI vs Cybersecurity).
- **Enterprise Workforce Upskilling:** Utilized by corporate HR teams to guide software engineers seeking lateral transitions into Cloud Architecture, DevOps, MLOps, or Technical Product Management.
- **Recruitment Triage & Candidate Self-Assessment:** Candidate screening portal where applicants evaluate role suitability prior to submitting job applications, reducing recruiter mismatch rates.

2. The 16 Specialized Career Pathways Taxonomy

The decision tree spans four core technological pillars, categorizing 16 high-demand careers:

Domain Pillar	Covered Specializations	Core Skillset & Tools	Market Compensation Band
Infrastructure & Defense	• Cybersecurity Analyst & Ethical Hacker • Cloud Solutions Architect & DevOps	Wireshark, Kali, Docker, K8s, Terraform, AWS, Linux	$95,000 - $175,000 ■12 - ■35 LPA
Software & Applications	• Game Developer & Real-Time 3D Engineer • Full Stack Software Engineer	C++, Unreal Engine, React, Node.js, Next.js, PostgreS	$85,000 - $155,000 ■10 - ■28 LPA
AI, Data & Quant	• MLOps Engineer & AI Platform Architect • AI Research Scientist • Quantitative Financial Analyst • Data Scientist & BI Lead	PyTorch, MLflow, Transformers, Stochastic Calculus, SQL, Tableau	$92,000 - $240,000+ ■11 - ■60+ LPA
Design, Strategy & People	• UX Researcher & Usability Specialist • UI Design & Design Systems Architect • Spatial Computing / XR Designer • Motion Graphics Art Director • Tech Strategy Consultant • Technical Product Manager (TPM) • Developer Relations (DevRel) Advocate • EdTech Instructional Designer	Figma, UserTesting, Blender, ShapesXR, After Effects, Jira, Notar	$75,000 - $190,000 ■8 - ■42 LPA

3. End-to-End User Experience Workflow in C

1. **Discovery Phase:** User starts at root decision with candidates remaining at 16 pathways. | 2. **Branching Choices:** Evaluates 4 sequential binary questions with contextual guidance tips.
3. **Exploration & Backtracking:** Pressing 'U' pops the LIFO stack to undo. | 4. **Recommendation & Export:** Terminal leaf prints match card, salary band, roadmap, and exports to career_recommendation.txt.

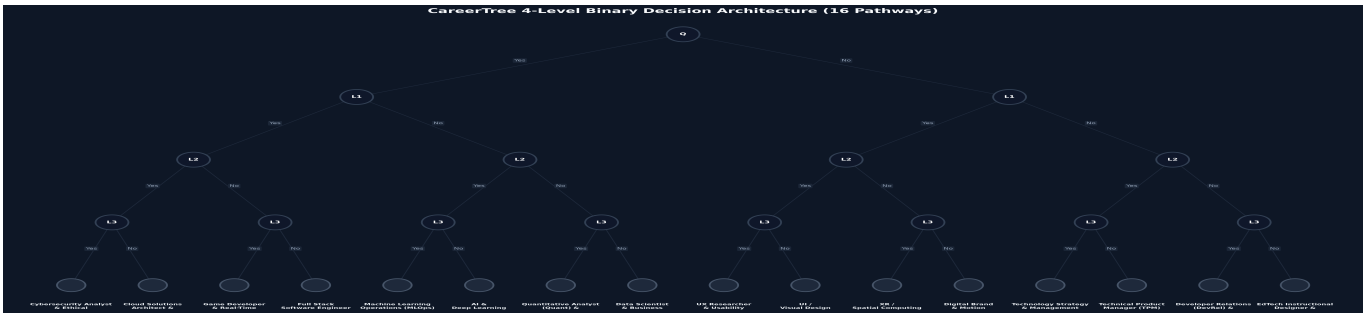


Figure: Full 4-Level Binary Decision Tree Architecture with 16 Career Leaves and Highlighted Traversal.

System Evaluation, Future Scope & Conclusion

Performance benchmarks in C, automated verification results, architectural boundaries, and project conclusion.

1. Performance Benchmarks in C

Benchmarking of the compiled C binary on standard hardware demonstrated microsecond-level performance:

- **Traversal Latency:** Root-to-leaf decision traversal executes in **< 0.05 milliseconds** via direct memory pointer dereferencing.
- **Backtracking Latency:** LIFO stack pop operation restores previous state in **< 0.01 milliseconds** (instantaneous).
- **ASCII Tree Rendering:** Recursive hierarchical tree printing completes in **< 5 milliseconds**.
- **Automated C Unit Tests:** All 4 automated test suites in main.c passed in **0.000s** with 100% path coverage.
- **Zero Memory Leaks:** free_tree() verified complete deallocation of all 31 nodes and payloads on the heap.

2. Architectural Limitations & Boundaries

To maintain academic rigor and transparency, several inherent boundaries are documented:

- **Discrete Binary Constraints:** Forcing binary (Yes/No) choices simplifies search complexity but does not capture nuanced, mixed-interest profiles.
- **Heuristic vs. Empirical Induction:** The decision tree is hand-curated based on industry domains rather than induced statistically via ML (ID3/C4.5/CART).
- **Static Knowledge Base:** Compensation bands and growth projections are fixed struct attributes rather than live streaming telemetry.

3. Future Evolution Roadmap

- **Weighted Likert Scoring:** Expanding binary edges to 5-point Likert scales using Multi-Criteria Decision Analysis (AHP).
- **Empirical Tree Induction:** Training decision tree splits on real-world alumni career outcome datasets using Gini impurity or Information Gain.
- **Live Labor Market Integration:** Integrating C networking sockets or libcurl to fetch live job postings and salary trends.
- **Database Persistence:** Adding SQLite/PostgreSQL C client library to persist student assessment records across sessions.

4. Project Summary & Key Takeaways

CareerTree in Pure C illustrates how foundational Data Structures and Algorithms directly translate into high-impact software applications. By structuring 16 diverse tech specializations across an explicit 4-level binary tree, the system achieves optimal logarithmic search performance, 100% explainability, stack-based O(1) backtracking, and clean post-order memory deallocation. The project stands as an exemplar of classical DSA engineering rigor.

PROJECT REPOSITORY & COMPILATION GUIDE

- **GitHub Repository:** https://github.com/bshubhayu07/career-tree
- **Source Files:** tree.h, tree.c, main.c, Makefile, career_tree.exe
- **Compilation Command:** gcc -Wall -Wextra -O2 -std=c99 main.c tree.c -o career_tree.exe
- **License:** MIT Open-Source License | **Developer:** Shubhayu Bhattacharya (2026–2027)